Ecosystem

Ch 12 ECOSYSTEM - Chapter Opener

Chapter contents (as printed on the opener page): 12.1 Ecosystem - Structure and Function; 12.2 Productivity; 12.3 Decomposition; 12.4 Energy Flow; 12.5 Ecological Pyramids.

- An **ecosystem** can be visualised as a **functional unit of nature**, where living organisms interact among themselves and also with the surrounding physical environment.
  - Ecosystem varies greatly in **size** - from a **small pond** to a **large forest or a sea**.
  - Many ecologists regard the **entire biosphere as a global ecosystem**, as a **composite of all local ecosystems on Earth**.

Since this system is **too much big and complex** to be studied at one time, it is convenient to divide it into **two basic categories**, namely **terrestrial** and **aquatic**.

Category	Examples (NCERT)
Terrestrial ecosystems	**Forest, grassland and desert**
Aquatic ecosystems	**Pond, lake, wetland, river and estuary**
Man-made ecosystems	**Crop fields and an aquarium** may also be considered as man-made ecosystems

We will first look at the **structure** of the ecosystem, in order to appreciate the **input (productivity)**, **transfer of energy (food chain/web, nutrient cycling)** and the **output (degradation and energy loss)**. We will also look at the relationships **- cycles, chains, webs -** that are created as a result of these energy flows within the system and their inter-relationship.

12.1 ECOSYSTEM - STRUCTURE AND FUNCTION

In earlier classes, you have looked at the various components of the environment **- abiotic** and **biotic**. You studied how the individual biotic and abiotic factors affected each other and their surrounding. Let us look at these components in a more **integrated manner** and see how the **flow of energy** takes place within these components of the ecosystem.

Interaction of biotic and abiotic components results in a **physical structure that is characteristic for each type of ecosystem**.

Component	What it comprises (Summary)
Abiotic components	Inorganic materials - **air, water and soil**
Biotic components	**Producers, consumers and decomposers**

- **Species composition**: identification and enumeration of plant and animal species of an ecosystem gives its species composition.
- **Stratification**: vertical distribution of different species occupying different levels is called stratification. For example, **trees** occupy the top vertical strata or layer of a forest, **shrubs** the second, and **herbs and grasses** occupy the bottom layers.

> *[NOTE] **Species composition** and **stratification** are the **two main structural features** of an ecosystem (stated explicitly in the Summary).*

The components of the ecosystem are seen to function as a unit when you consider the following **four aspects**, in NCERT's order:

1. **Productivity**
2. **Decomposition**
3. **Energy flow**
4. **Nutrient cycling**

12.1 The pond as an example of an aquatic ecosystem

To understand the ethos of an aquatic ecosystem let us take a **small pond** as an example. This is **fairly a self-sustainable unit** and rather simple example that explains even the complex interactions that exist in an aquatic ecosystem. A pond is a **shallow water body** in which all the above-mentioned **four basic components** of an ecosystem are well exhibited.

Component in the pond	What it is
Abiotic	The **water** with all the **dissolved inorganic and organic substances** and the **rich soil deposit** at the bottom of the pond. The **solar input**, the **cycle of temperature, day-length** and other climatic conditions regulate the rate of function of the entire pond
Autotrophic components	The **phytoplankton**, some **algae** and the **floating, submerged and marginal plants** found at the edges
Consumers	The **zooplankton**, the **free swimming and bottom dwelling forms**
Decomposers	The **fungi, bacteria and flagellates** especially abundant in the bottom of the pond

This system performs **all the functions of any ecosystem and of the biosphere as a whole**, i.e., **conversion of inorganic into organic material** with the help of the radiant energy of the sun by the **autotrophs**; **consumption of the autotrophs by heterotrophs**; **decomposition and mineralisation of the dead matter** to release them back for reuse by the autotrophs. These events are **repeated over and over again**.

- There is a **unidirectional movement of energy** towards the higher trophic levels and its **dissipation and loss as heat** to the environment.

12.2 PRODUCTIVITY

A **constant input of solar energy** is the **basic requirement** for any ecosystem to function and sustain.

- **Primary production** is defined as the **amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis**. It is expressed in terms of **weight ($g\ m^{-2}$)** or **energy ($kcal\ m^{-2}$)**.
- The **rate of biomass production** is called **productivity**. It is expressed in terms of **$g\ m^{-2}\ yr^{-1}$** or **$(kcal\ m^{-2})\ yr^{-1}$** to compare the productivity of different ecosystems.

Productivity can be divided into **gross primary productivity (GPP)** and **net primary productivity (NPP)**.

Term	Definition
Gross primary productivity (GPP)	The **rate of production of organic matter during photosynthesis**. A considerable amount of GPP is **utilised by plants in respiration**
Net primary productivity (NPP)	**Gross primary productivity minus respiration losses (R)**, i.e. **GPP - R = NPP**. NPP is the **available biomass for the consumption to heterotrophs (herbivores and decomposers)**
Secondary productivity	Defined as the **rate of formation of new organic matter by consumers**

> ***[NOTE]*** *NCERT's parenthetical spells it **"herbiviores"**; the fact is that NPP is available to **herbivores AND decomposers**. The Summary also states secondary productivity as the **rate of assimilation of food energy by the consumers** - recognise both wordings.*

Primary productivity depends on the **plant species** inhabiting a particular area. It also depends on a variety of **environmental factors**, **availability of nutrients** and **photosynthetic capacity of plants**. Therefore, it **varies in different types of ecosystems**.

Annual net primary productivity	Value (dry weight of organic matter)
Whole biosphere	Approximately **170 billion tons**
Oceans (despite occupying about **70 per cent** of the surface)	Only **55 billion tons**
Land	The **rest**, of course, is on land

> ***[NOTE] NCERT activity:*** *Discuss the **main reason for the low productivity of ocean** with your teacher.*

12.3 DECOMPOSITION

You may have heard of the **earthworm** being referred to as the **farmer's 'friend'**. This is so because they help in the **breakdown of complex organic matter** as well as in **loosening of the soil**.

- **Decomposition**: decomposers break down complex organic matter into **inorganic substances** like **carbon dioxide, water and nutrients**, and the process is called decomposition.
- **Detritus**: dead plant remains such as **leaves, bark, flowers** and **dead remains of animals, including fecal matter**, constitute detritus, which is the **raw material for decomposition**.

The important steps in the process of decomposition are (in NCERT's order):

1. **Fragmentation - detritivores (e.g., earthworm)** break down detritus into **smaller particles**. (F060)
2. **Leaching - water-soluble inorganic nutrients** go down into the **soil horizon** and get **precipitated as unavailable salts**. (F061)
3. **Catabolism - bacterial and fungal enzymes degrade detritus into simpler inorganic substances**. (F062)
4. **Humification -** leads to accumulation of **humus** (see below). (F064-F066)
5. **Mineralisation -** the humus is further degraded to release inorganic nutrients (see below). (F067)

> *[NOTE] It is important to note that **all the above steps in decomposition operate simultaneously** on the detritus (Figure 12.1). Humification and mineralisation occur **during decomposition in the soil**.*

- **Humification** leads to accumulation of a **dark coloured amorphous substance called humus** that is **highly resistant to microbial action** and **undergoes decomposition at an extremely slow rate**. Being **colloidal in nature** it serves as a **reservoir of nutrients**.
- **Mineralisation**: the humus is **further degraded by some microbes** and **release of inorganic nutrients** occurs by this process.

> ***[NOTE] Five vs three - a discrepancy to know:** the **body lists FIVE steps** (fragmentation, leaching, catabolism, humification and mineralisation), but the **Summary says decomposition involves THREE processes - fragmentation of detritus, leaching and catabolism**. Both wordings are NCERT's; do not reconcile them - recognise either in a question.*

Decomposition is **largely an oxygen-requiring process**. The **rate of decomposition** is controlled by **chemical composition of detritus** and **climatic factors**:

Factor	Effect on the rate of decomposition
Chemical composition of detritus	In a particular climatic condition, decomposition is **slower** if detritus is rich in **lignin and chitin**, and **quicker** if detritus is rich in **nitrogen and water-soluble substances like sugars**
Climatic factors (temperature and soil moisture)	The **most important** climatic factors; they regulate decomposition through their effects on the **activities of soil microbes**. **Warm and moist environment favour decomposition**, whereas **low temperature and anaerobiosis inhibit decomposition**, resulting in **build up of organic materials**

The decomposition cycle in a terrestrial ecosystem (Figure 12.1) reads as follows: a tree grows in the soil; **a green leaf falls to the ground**; **some are eaten by insects and other animals**, so that **nutrients and energy enter the food web**; **some nutrients leach into soil by chemical action**; **leaves partially consumed by decomposers such as fungi and bacteria begin to lose form and become litter**; there is **further decomposition by earthworms, bacteria, soil mites, fungi, etc.**; and the end result is **organic rich soil**.

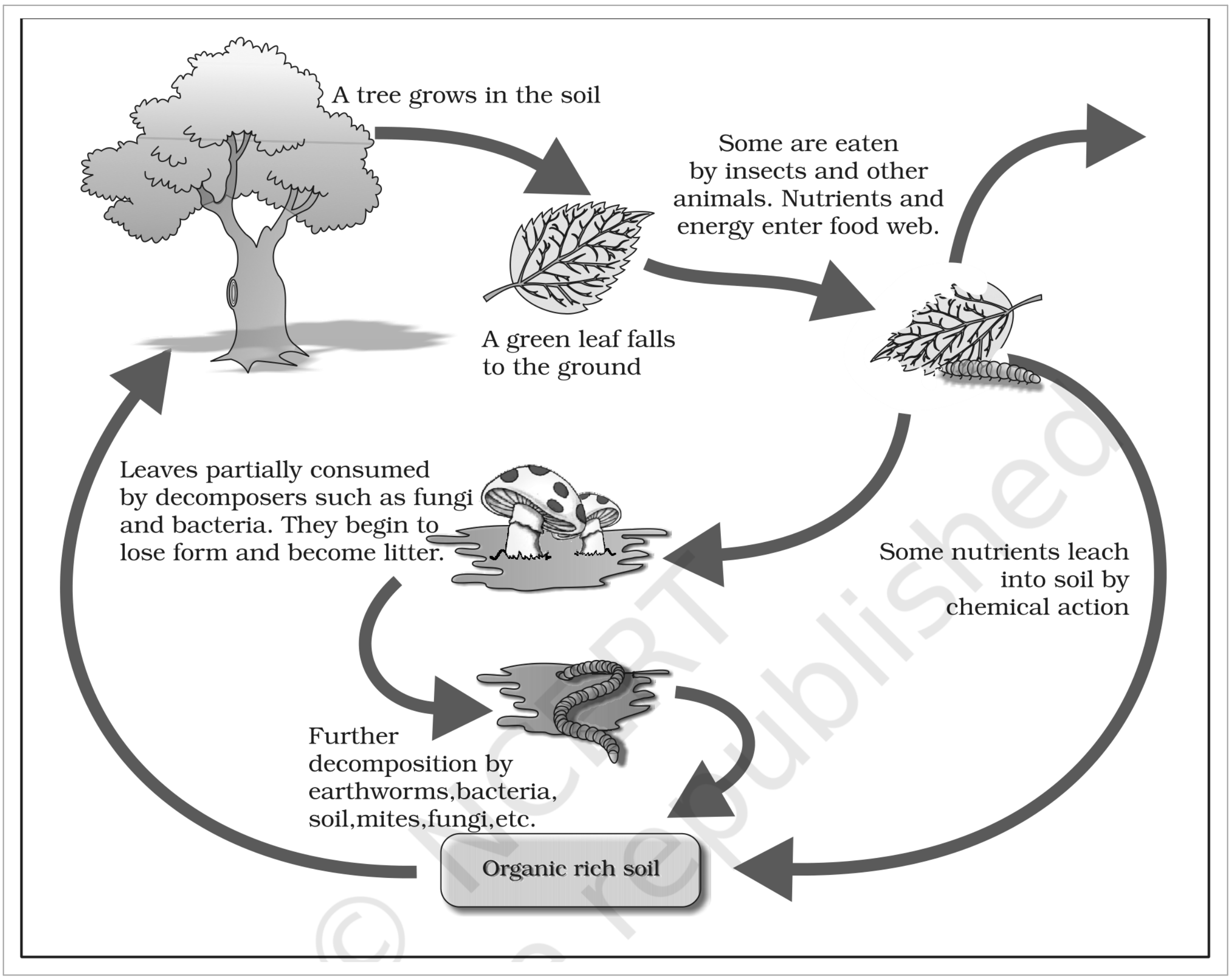

Fig. 12.1 - Diagrammatic representation of decomposition cycle in a terrestrial ecosystem.

12.4 ENERGY FLOW

- **Except for the deep sea hydro-thermal ecosystem**, the **sun is the only source of energy for all ecosystems on Earth**. Of the incident solar radiation, **less than 50 per cent** of it is **photosynthetically active radiation (PAR)**.

Plants and photosynthetic bacteria (autotrophs) fix the Sun's radiant energy to **make food from simple inorganic materials**. Plants capture only **2-10 per cent of the PAR**, and this **small amount of energy sustains the entire living world**.

So it is very important to know how the solar energy captured by plants flows through different organisms of an ecosystem. All organisms are **dependent for their food on producers**, either **directly or indirectly**. So you find a **unidirectional flow of energy** from the sun to producers and then to consumers.

> *[NOTE] **Thermodynamics:** Is this in keeping with the **first law of thermodynamics**? Ecosystems are **not exempt from the Second Law of thermodynamics** either - they need a **constant supply of energy** to synthesise the molecules they require, to **counteract the universal tendency toward increasing disorderliness**.*

- **Producers**: the **green plants in the ecosystem** are called producers. In a **terrestrial ecosystem**, major producers are **herbaceous and woody plants**. In an **aquatic ecosystem** producers are various species like **phytoplankton, algae and higher plants**.

Starting from the **plants (or producers)**, **food chains** or rather **webs** are formed such that an **animal feeds on a plant or on another animal and in turn is food for another**. The chain or web is formed because of this **interdependency**.

- **No energy that is trapped into an organism remains in it for ever.** The energy trapped by the producer is **either passed on to a consumer or the organism dies**.
- **Death of an organism is the beginning of the detritus food chain/web.**

- **Consumers (heterotrophs)**: all animals depend on plants (directly or indirectly) for their food needs, hence they are called **consumers** and also **heterotrophs**.

Consumer category	Definition (NCERT)
Primary consumers	Animals that **feed on the producers (the plants)**. Obviously the primary consumers will be **herbivores**
Secondary consumers	Animals that **eat other animals which in turn eat the plants (or their produce)**. You could have **tertiary consumers** too
Primary carnivores	The consumers that **feed on herbivores** are carnivores, or more correctly **primary carnivores (though secondary consumers)**
Secondary carnivores	Those animals that **depend on the primary carnivores for food**

- Some common **herbivores** are **insects, birds and mammals** in the terrestrial ecosystem and **molluscs** in the aquatic ecosystem.

A **simple grazing food chain (GFC)** is depicted below:

1. **Grass (Producer)**
2. **Goat (Primary Consumer)**
3. **Man (Secondary consumer)**

- **Detritus food chain (DFC)** begins with **dead organic matter**. It is made up of **decomposers**, which are **heterotrophic organisms, mainly fungi and bacteria**. They meet their energy and nutrient requirements by **degrading dead organic matter or detritus**. These are also known as **saprotrophs** (**sapro**: to decompose).
  - Decomposers **secrete digestive enzymes** that break down dead and waste materials into **simple, inorganic materials**, which are subsequently **absorbed by them**.

Ecosystem	Dominant energy-flow route
Aquatic ecosystem	**GFC is the major conduit for energy flow**
Terrestrial ecosystem	A **much larger fraction of energy** flows through the **detritus food chain** than through the GFC

The **detritus food chain may be connected with the grazing food chain** at some levels: **some of the organisms of DFC are prey to the GFC animals**, and in a natural ecosystem **some animals like cockroaches, crows, etc., are omnivores**. These **natural interconnections of food chains make it a food web**.

> ***[NOTE] NCERT prompt:*** *How would you classify human beings!*

- **Trophic level**: organisms occupy a place in the community according to their **feeding relationship** with other organisms. Based on the **source of their nutrition or food**, organisms occupy a specific place in the food chain that is known as their trophic level.

Trophic levels in an ecosystem (Figure 12.2): producers belong to the **first trophic level**, herbivores (primary consumer) to the **second**, and carnivores (secondary consumer) to the **third**. The important point to note is that the **amount of energy decreases at successive trophic levels**.

Trophic level	Category	Organism type	Examples
First trophic level	**Producer**	**Plants**	**Phytoplankton, grass, trees**
Second trophic level	**Primary Consumer**	**Herbivore**	**Zooplankton, grasshopper and cow**
Third trophic level	**Secondary Consumer**	**Carnivore**	**Birds, fishes wolf**
Fourth Trophic level	**Tertiary Consumer**	**Top Carnivore**	**Man, lion**

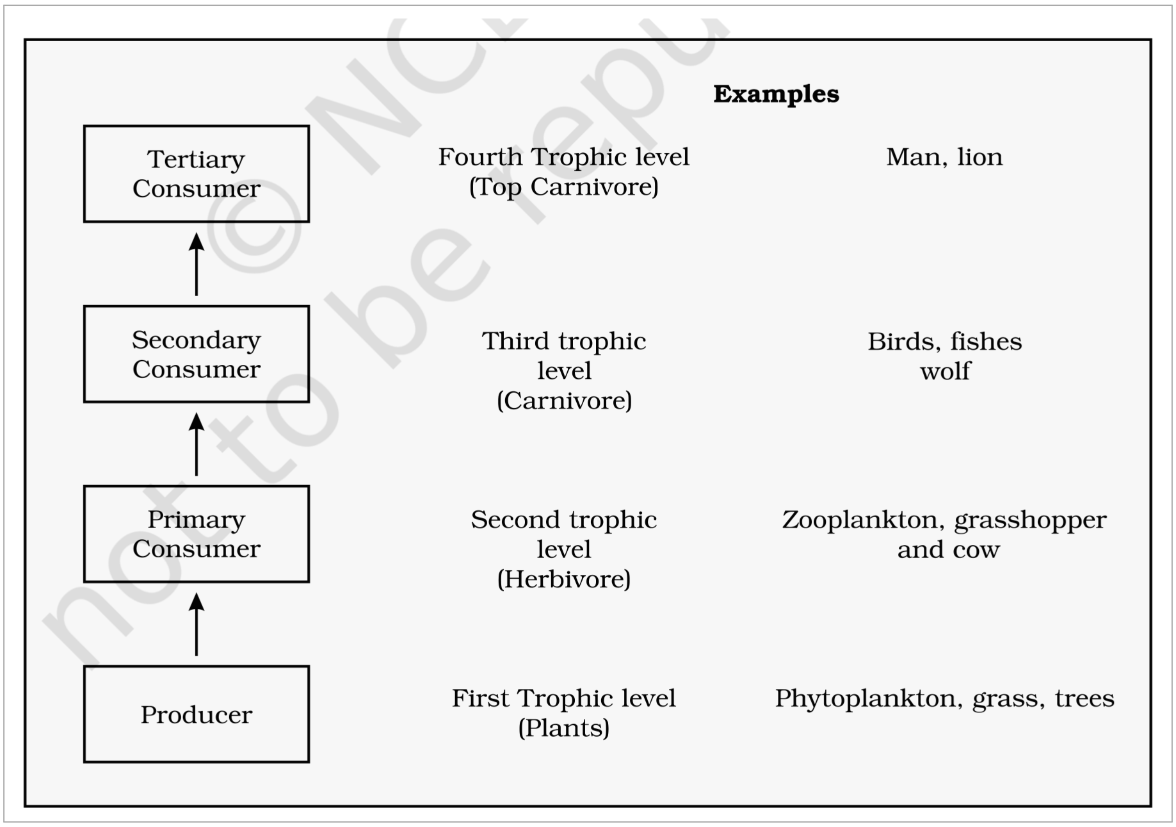

Fig. 12.2 - Diagrammatic representation of trophic levels in an ecosystem.

When any organism dies it is **converted to detritus or dead biomass** that serves as an **energy source for decomposers**. Organisms at each trophic level depend on those at the **lower trophic level** for their energy demands.

- **Standing crop**: each trophic level has a certain **mass of living material at a particular time** called the standing crop. It is measured as the **mass of living organisms (biomass)** or the **number in a unit area**.
  - The **biomass** of a species is expressed in terms of **fresh or dry weight**. Measurement of biomass in terms of **dry weight is more accurate**. (**NCERT prompt:** Why?)
- **10 per cent law:** the number of trophic levels in the grazing food chain is **restricted as the transfer of energy follows the 10 per cent law -** only **10 per cent** of the energy is transferred to each trophic level from the lower trophic level.

In nature, it is possible to have **so many levels - producer, herbivore, primary carnivore, secondary carnivore -** in the grazing food chain (Figure 12.3). (**NCERT prompt:** Do you think there is any such limitation in a detritus food chain?)

Energy flow through different trophic levels (Figure 12.3): energy from the **Sun** enters the **first trophic level producers (plants)**, passes to the **second trophic level primary consumers (herbivores)**, then to the **third trophic level secondary consumers (carnivores)** and the **fourth trophic level tertiary consumers (top carnivores)**; at every step some energy is lost as **Heat**.

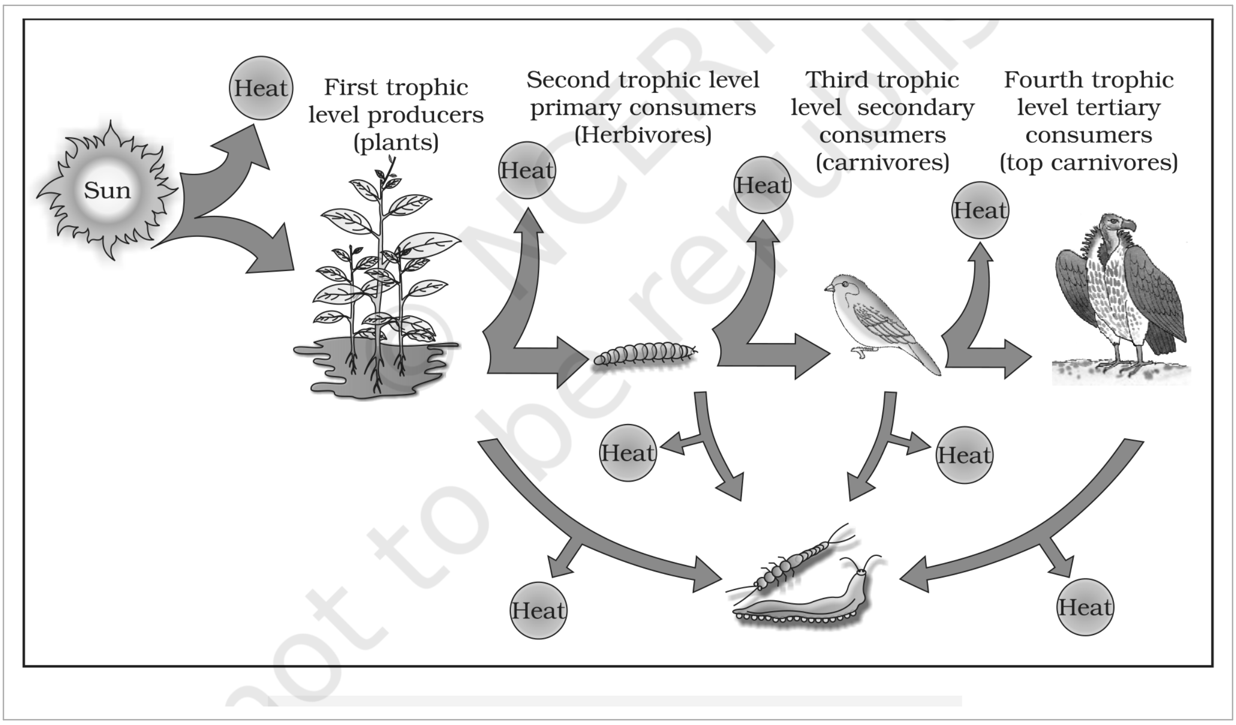

Fig. 12.3 - Energy flow through different trophic levels. The original colour figure separates the trophic levels by hue; after monochrome conversion they are read by their position, labels and the Sun/Heat markers.

12.5 ECOLOGICAL PYRAMIDS

You must be familiar with the **shape of a pyramid**: the **base is broad and it narrows towards the apex**. One gets a similar shape whether you express the **food or energy relationship** between organisms at different trophic levels. This relationship is expressed in terms of **number, biomass or energy**.

- The **base** of each pyramid represents the **producers or the first trophic level**, while the **apex** represents the **tertiary or top level consumer**.

The **three types of ecological pyramids** that are usually studied are **(a) pyramid of number**; **(b) pyramid of biomass**; and **(c) pyramid of energy** (for detail see Figure 12.4 a, b, c and d).

- Any calculations of **energy content, biomass or numbers** has to **include all organisms at that trophic level**. No generalisations we make will be true if we take **only a few individuals** at any trophic level into account.
- A given organism **may occupy more than one trophic level simultaneously**. One must remember that the **trophic level represents a functional level, not a species as such**.
- For example, a **sparrow** is a **primary consumer** when it eats **seeds, fruits, peas**, and a **secondary consumer** when it eats **insects and worms**. (**NCERT prompt:** Can you work out how many trophic levels human beings function at in a food chain?)

In **most ecosystems**, all the pyramids **- of number, of energy and biomass -** are **upright**, i.e., **producers are more in number and biomass than the herbivores**, and **herbivores are more in number and biomass than the carnivores**. Also, **energy at a lower trophic level is always more than at a higher level**.

The values shown in the four pyramids of Figure 12.4 (every bar value carried verbatim; P = Producer, PC = Primary consumer, SC = Secondary consumer, TC = Tertiary consumer):

Trophic level	(a) Pyramid of numbers, grassland: Number of individuals	(b) Pyramid of biomass, grassland: Dry weight (kg m^{-2})	(c) Inverted pyramid of biomass (aquatic)	(d) Pyramid of energy
P (Producer)	**5,842,000**	**809**	**4**	**10,000 J**
PC (Primary consumer)	**708,000**	**37**	**21**	**1000 J**
SC (Secondary consumer)	**3,54,000**	**11**	**-**	**100 J**
TC (Tertiary consumer)	**3**	**1.5**	**-**	**10 J**

ⓘ ***[NOTE] Figure 12.4 caption facts:*** *(a) Only* ***three top-carnivores*** *are supported in an ecosystem based on production of* ***nearly 6 millions plants -*** *the bar itself reads* ***5,842,000*** *(carry both, do not reconcile). (d) An* ***ideal pyramid of energy****: primary producers convert* ***only 1% of the energy in the sunlight*** *available to them into NPP, out of* ***1,000,000 J of Sunlight****.*

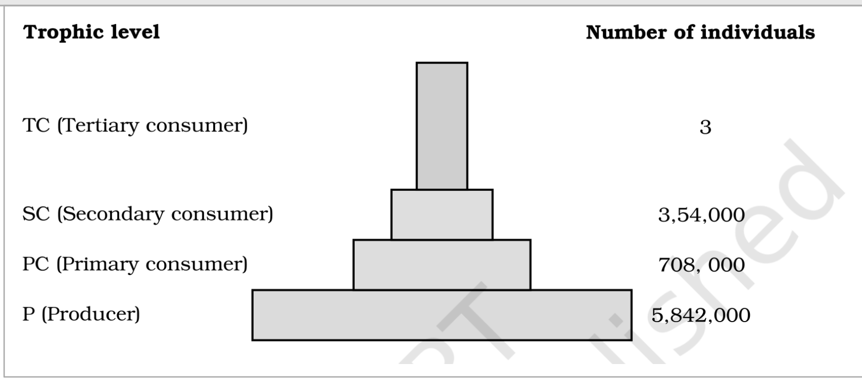

Fig. 12.4 (a) - Pyramid of numbers in a grassland ecosystem. Only three top-carnivores are supported in an ecosystem based on production of nearly 6 millions plants.

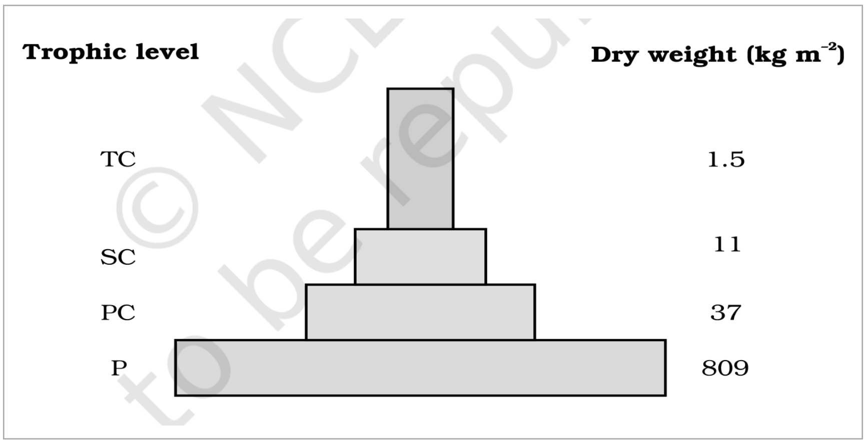

Fig. 12.4 (b) - Pyramid of biomass shows a sharp decrease in biomass at higher trophic levels.

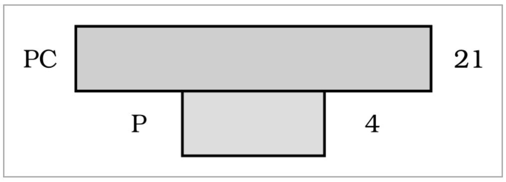

Fig. 12.4 (c) - Inverted pyramid of biomass - small standing crop of phytoplankton supports large standing crop of zooplankton.

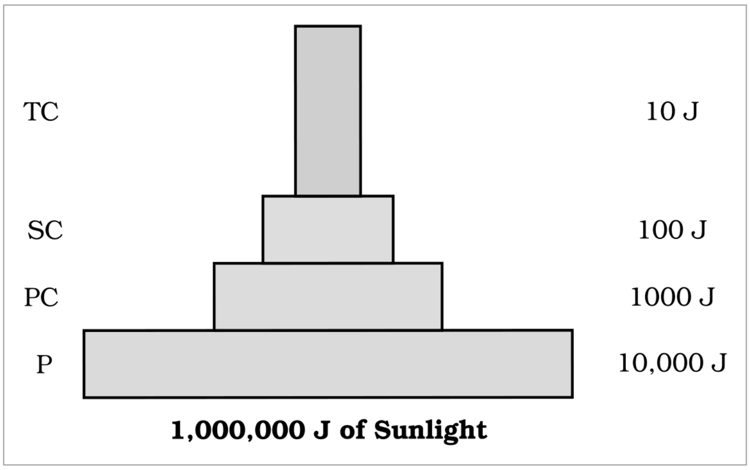

Fig. 12.4 (d) - An ideal pyramid of energy. Observe that primary producers convert only 1% of the energy in the sunlight available to them into NPP.

There are exceptions to the "pyramids are upright" generalisation:

- **Pyramid of numbers on a big tree:** if you count the number of **insects feeding on a big tree**, then add the **small birds depending on the insects** and the **larger birds eating the smaller**, the pyramid of numbers you get is **not upright** (it is inverted/spindle-shaped). (NCERT asks you to draw it.)
- **Pyramid of biomass in the sea** is **generally inverted** because the **biomass of fishes far exceeds that of phytoplankton**. (**NCERT prompt:** Isn't that a paradox? How would you explain this?)

- **Pyramid of energy is always upright, can never be inverted**, because when energy flows from a particular trophic level to the next, **some energy is always lost as heat at each step**.
- Each bar in the **energy pyramid** indicates the **amount of energy present at each trophic level in a given time or annually per unit area**.

> ⓘ ***[NOTE] Limitations of ecological pyramids:*** *(1) it* ***does not take into account the same species belonging to two or more trophic levels****; (2) it* ***assumes a simple food chain****, something that* ***almost never exists in nature****, and* ***does not accommodate a food web****; (3)* ***saprophytes are not given any place*** *in ecological pyramids even though they play a* ***vital role in the ecosystem****.*

12.6 NUTRIENT CYCLING AND ECOSYSTEM SERVICES

> ⓘ ***[NOTE]*** *This section carries facts stated* ***only in the NCERT Summary*** *- nutrient cycling is listed as one of the four functions in 12.1 but is never defined in the body, so it is folded in here.*

- **Nutrient cycling**: the **storage and movement of nutrient elements through the various components of the ecosystem** is called nutrient cycling; **nutrients are repeatedly used** through this process.

Type of nutrient cycle	Reservoir (Summary)
Gaseous type (e.g., **carbon**)	**Atmosphere or hydrosphere**
Sedimentary type (e.g., **phosphorus**)	**Earth's crust**

- **Ecosystem services**: the **products of ecosystem processes** are named ecosystem services - e.g., **purification of air and water by forests**.

Recap QUICK RECAP

- An **ecosystem** is a **structural and functional unit of nature** comprising **abiotic** components (inorganic materials - **air, water, soil**) and **biotic** components (**producers, consumers, decomposers**). Each ecosystem has a **characteristic physical structure** from the interaction of these components.
- **Species composition** and **stratification** are the **two main structural features** of an ecosystem. Based on the source of nutrition, every organism occupies a place in an ecosystem.
- **Productivity, decomposition, energy flow and nutrient cycling** are the **four important functions/components** of an ecosystem.
- **Primary productivity** is the **rate of capture of solar energy or biomass production of the producers**. **GPP** is the total production of organic matter; **NPP** is the **remaining biomass/energy left after utilisation by producers** (GPP - R). **Secondary productivity** is the **rate of assimilation of food energy by the consumers**.
- In **decomposition**, complex organic compounds of **detritus** are converted to **carbon dioxide, water and inorganic nutrients** by the decomposers. The body lists **five** steps; the Summary names **three - fragmentation, leaching and catabolism**.
- **Energy flow is unidirectional**: first, plants capture solar energy, and then food is transferred from the **producers to decomposers**. Organisms of different trophic levels are connected for food/energy, forming a **food chain** and **food web**.
- **Nutrient cycling** is the storage and movement of nutrients through the ecosystem; it is of **two types - gaseous** (reservoir: atmosphere/hydrosphere, e.g. carbon) and **sedimentary** (reservoir: Earth's crust, e.g. phosphorus). **Ecosystem services** are the products of ecosystem processes, e.g. **purification of air and water by forests**.

Appendix TERMS USED IN THE EXERCISES

NCERT's eleven exercise questions assume a few terms/facts the chapter never states outright. Everything below is built only from statements already made in this chapter (plus the Summary).

Term / fact assumed	Explanation (from this chapter)
Litter (Ex 6(e): distinguish **litter and detritus**)	The body never defines litter; it appears only in Figure 12.1. **Litter** = freshly fallen, still-recognisable dead plant/animal material on the soil surface, which - as decomposers begin to consume it - loses form and becomes part of the **detritus** (all dead plant/animal remains that are the raw material for decomposition)
Limiting factor for productivity in aquatic ecosystems (Ex 1(c))	The body only asks you to discuss low ocean productivity. The **limiting factor is light (and nutrient) availability** in water, tied to the ocean's low 55-billion-ton productivity despite its large area
Major reservoir of carbon on earth (Ex 1(e))	From the nutrient-cycling block: **carbon** follows a **gaseous** cycle whose reservoir is the **atmosphere or hydrosphere**
Secondary producers (Ex 4)	The term appears nowhere in the chapter; NCERT uses **secondary productivity** and **primary/secondary consumers**. As there is no such category, the intended answer is **(d) None of the above**
Pyramid of numbers in a tree-dominated ecosystem (Ex 1(b))	Posed as an activity (F146). The pyramid is **inverted / spindle-shaped**, the same non-upright case as the big-tree/insects/birds example

Appendix NCERT Exercises (for reference)

1. Fill in the blanks: (a) Plants are called **autotrophs/producers** because they fix carbon dioxide; (b) in an ecosystem dominated by trees, the pyramid of numbers is **inverted/spindle** type; (c) in aquatic ecosystems, the limiting factor for productivity is **light**; (d) common detritivores are **earthworms**; (e) the major reservoir of carbon on earth is **oceans/hydrosphere**.

2-5 (MCQ): 2. Largest population in a food chain **- (a) Producers**. 3. The second trophic level in a lake **- (b) Zooplankton**. 4. Secondary producers are **- (d) None of the above**. 5. Percentage of PAR in incident solar radiation **- (b) 50%** (NCERT states "less than 50 per cent").

6. Distinguish between: (a) Grazing food chain and detritus food chain; (b) Production and decomposition; (c) Upright and inverted pyramid; (d) Food chain and Food web; (e) Litter and detritus; (f) Primary and secondary productivity **-** all covered above.

7-11 (long answers): 7. Describe the components of an ecosystem. 8. Define ecological pyramids and describe, with examples, pyramids of number and biomass. 9. What is primary productivity? Describe factors that affect it. 10. Define decomposition and describe its processes and products. 11. Give an account of energy flow in an ecosystem **-** all answered by the sections above.

Every fact, number, name, qualifier, table row, figure and figure label in NCERT Class 12 Chapter 12 is carried above. Nothing outside the source chapter has been added.